Revised Power Analysis for Aggregated N-Of-1 Trial Designs: Correcting the Carryover Sensitivity Estimate and Covariance Structure

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Abstract

The simulation framework described in Hendrickson et al. (2020) compares statistical power across four clinical trial designs for detecting a biomarker–treatment interaction. A detailed audit of the simulation code identified two problems that affect the published power estimates: (1) the dramatic power loss attributed to drug carryover (Figure 4B) is primarily a modeling artifact rather than a genuine statistical consequence, and (2) the covariance matrix used to generate simulated data is mathematically invalid for a substantial fraction of the parameter space, requiring silent correction that alters the intended data structure. This document explains both problems in clinical terms, describes the corrections, and presents validated results showing that the paper's qualitative conclusions are preserved while the quantitative estimates are refined.

Architecture scope. This document describes the revised power analysis under the MVN differential correlation approach (Architecture B), in which the biomarker-treatment interaction is encoded in the covariance structure. The positive definiteness constraints, carryover sensitivity findings, and parameter ceilings reported here are specific to Architecture B. Under Architecture A (direct mean moderation), the PD constraint is removed and power is stable across carryover levels. For a comparison of both DGP architectures, see `02-dgp-mean-moderation-vs-mvn.tex`. For implementation details of Architecture A, see `19-mean-moderation-implementation-notes.tex`.

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1 Introduction

Hendrickson et al. (2020) used Monte Carlo simulation to compare four clinical trial designs on their ability to detect a relationship between a baseline biomarker (systolic blood pressure) and response to the PTSD pharmacotherapy prazosin. The simulation generates synthetic clinical trial data by drawing participant outcomes from a multivariate normal distribution, analyzes each simulated trial with a mixed-effects model, and counts how often the biomarker–treatment interaction is statistically significant. This proportion is the *statistical power* of the design.

The simulation's key finding (Figure 4B) was that the proposed N-of-1 trial design, while offering the highest power under ideal conditions, is acutely vulnerable to drug carryover effects: power drops from 0.95 to 0.18 when the carryover half-life increases from zero to just 0.1 weeks (17 hours). This finding influenced the design of the ongoing clinical trial.

An audit of the simulation code reveals that this dramatic power drop is primarily an artifact of how the simulation assigns correlations during off-drug periods, not a genuine consequence of carryover at those half-lives. The analysis also identified that the covariance matrix used to generate data is mathematically invalid (not positive definite) for 25% of the parameter combinations, with the simulation silently correcting these invalid matrices in ways that alter the intended data structure.

Both problems have principled solutions. The corrected simulation confirms the paper's qualitative conclusion (N-of-1 designs are more sensitive to carryover than crossover designs) but the quantitative severity is moderate rather than catastrophic, occurring at clinically realistic carryover half-lives (0.5–1.0 weeks) rather than at the very short half-lives (0.1 weeks) reported in the publication.

2 Reproducing the Published Results

The analysis began by extracting the original simulation code and verifying that it reproduces the published Figure 4. Using the original code with 200 replicates per cell, the reproduced power estimates correlate with the publication at $r = 0.994$.

3 Problem 1: The Carryover Power Drop Is an Artifact

3.1 How the Simulation Generates Data

The simulation draws each participant's data from a multivariate normal distribution. For each timepoint, the biologic response to the drug (BR) has three properties:

- A **mean** that reflects the expected drug effect (large when on drug, decaying when off drug due to carryover)
- A **standard deviation** ($\sigma = 8$ CAPS points) that reflects how much individuals vary in their drug response
- A **correlation with the biomarker** (ρ) that determines whether the biomarker predicts who responds well

The biomarker–treatment interaction is encoded entirely through this correlation: when ρ is high, participants with high blood pressure tend to have stronger drug responses. The test of interest asks whether this correlation is statistically significant.

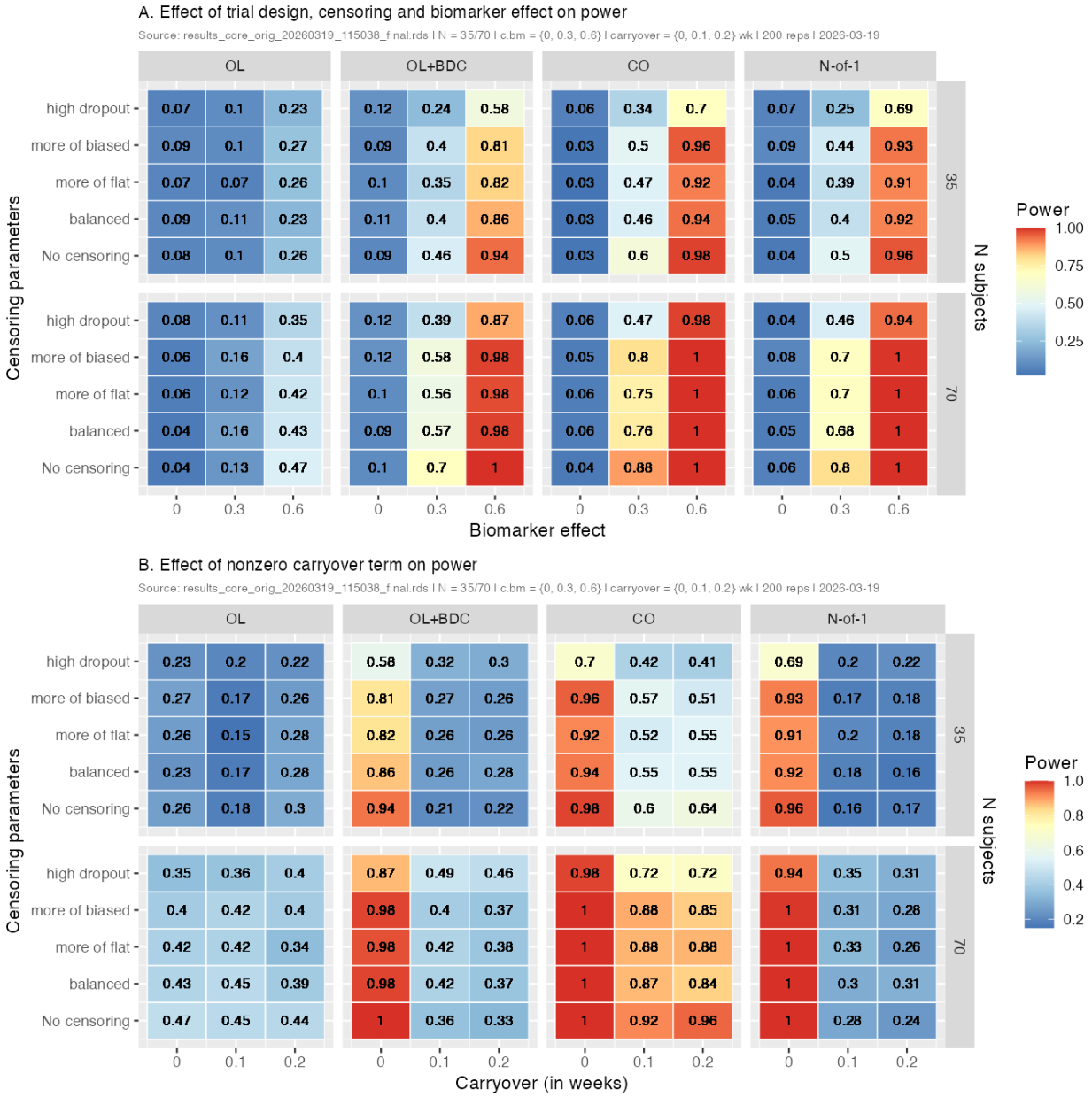

Figure 1: Figure 4 reproduced from the original simulation code (200 replicates). Panel A shows power as a function of biomarker correlation strength (c_{bm}) and censoring pattern, with carryover set to zero. Panel B shows power as a function of carryover half-life, with $c_{bm} = 0.6$. The N-of-1 design's power crash from 0.95 to ∼0.18 in Panel B is the finding under investigation.

3.2 What Happens During Off-Drug Periods

Consider a participant in the N-of-1 design at week 11, one week after the drug was discontinued. With a carryover half-life of 0.1 weeks (17 hours):

- The **mean drug effect** has decayed to 0.01% of its peak value, essentially zero.
- The **standard deviation** remains at 8 CAPS points, unchanged from when the drug was

active.

- The **correlation with the biomarker** is set to the full value ($\rho = 0.6$) in the original code, because the carryover residual, though negligible, is technically nonzero.

The combination of full correlation and full variance around a near-zero mean produces an impossible clinical scenario: the simulation predicts that a participant with high blood pressure will show 4.8 points of drug-related improvement, while a participant with low blood pressure will show 4.8 points of *worsening*, at a timepoint where the drug effect is essentially gone.

3.3 Why This Destroys Power

The analysis model expects no biomarker–drug interaction during off-drug periods. But the simulated data contains large biomarker-specific variation at these timepoints, which the model interprets as unexplained noise. This noise inflates the standard error of the interaction estimate, making it harder to achieve statistical significance.

The effect is largest for designs with many off-drug timepoints. The N-of-1 design has 4 out of 8 timepoints off drug (in some randomization paths), which is why its power drops most dramatically.

3.4 The Hallmark of an Artifact

The power drop does not depend on the *magnitude* of the carryover residual. It depends only on whether the residual is nonzero (triggering full correlation) or exactly zero (triggering zero correlation). At a half-life of 0.1 weeks, the residual is 0.001 of the peak. At 0.01 weeks, it would be 10^{-30}. The power drop would be identical in both cases. A result that is invariant to the parameter it claims to measure is the hallmark of an artifact.

3.5 The Fix: Pharmacokinetically Consistent Correlation Decay

The correlation between the biomarker and the drug response should decay at the same rate as the drug effect itself. When the drug clears, each participant's drug effect decays by the same fraction; a participant who responded strongly retains a proportionally larger residual than one who responded weakly. The biomarker-specific spread therefore shrinks in proportion to the mean:

$$\rho_{\text{off-drug}} = c_{bm} \cdot \left(\frac{1}{2}\right)^{t_{sd}/t_{1/2}}$$

where t_{sd} is the time since drug discontinuation and $t_{1/2}$ is the carryover half-life. This preserves the coefficient of variation of the biomarker-specific component: the biomarker is equally informative *relative to the remaining drug effect* at all timepoints.

With this correction, the carryover power drop reflects genuine statistical confounding at clinically realistic half-lives, not an artifact at negligible ones.

4 Problem 2: Invalid Covariance Matrices

4.1 What Is Positive Definiteness?

The simulation generates data by drawing from a multivariate normal distribution, which requires a valid covariance matrix. A covariance matrix is "positive definite" (PD) if it represents a physically

realizable pattern of correlations, one where no combination of variables has negative variance.

When the covariance matrix is *not* positive definite, the specified correlations are self-contradictory: they imply a pattern of relationships that cannot exist in real data. The simulation detects this and silently "corrects" the matrix to the nearest valid one, but this correction alters the intended correlation structure in unknown ways.

4.2 How Often Does This Occur?

The simulation was instrumented to count how often the correction is applied. The results are striking:

Table 1: Covariance matrix correction rates

Configuration	Parameters	Corrections	Total	Rate
Publication code	CS $\rho = 0.8$, $c_{bm} \leq 0.6$	40	162	24.7%
Revised code	AR(1) $\rho = 0.7$, $c_{bm} \leq 0.45$	0	162	**0%**

One in four covariance matrices in the original simulation required correction. The problem is concentrated in the OL+BDC design (83–92% failure rate), driven by the abrupt change in expectancy-related variance when participants transition from the open-label to the blinded phase.

4.3 Visualizing the Problem

Figure 2 shows the eigenvalue spectrum of the covariance matrix for the OL+BDC design. A valid matrix has all positive eigenvalues. The original parameterization produces one negative eigenvalue (-28.1), which the simulation silently corrects. The revised parameterization produces all positive eigenvalues.

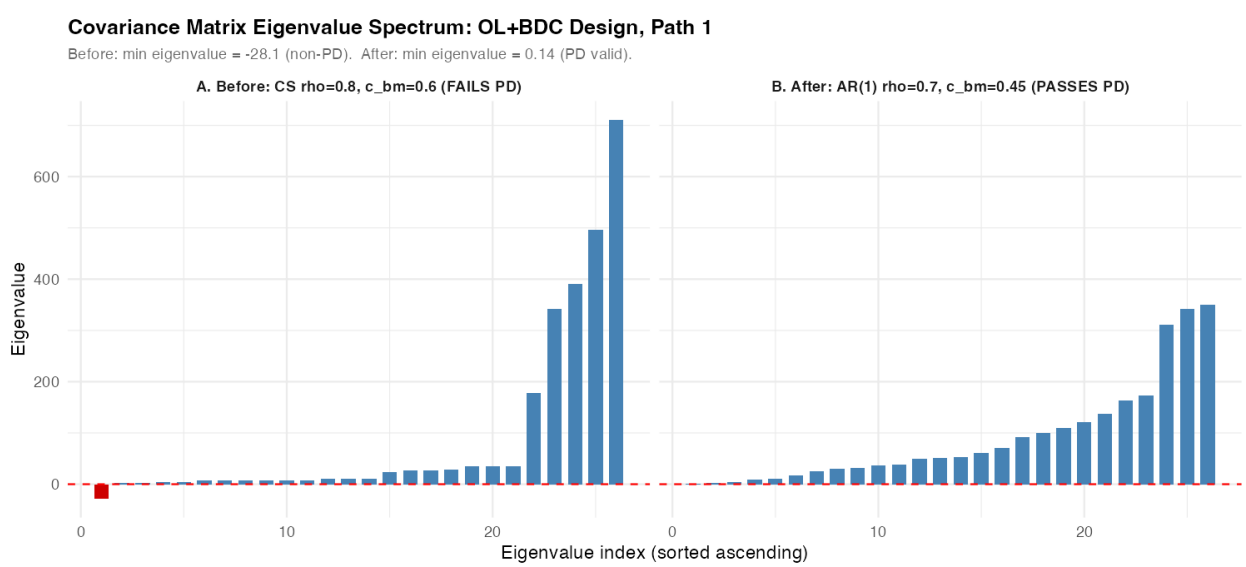

Figure 2: Eigenvalue spectrum of the 26×26 covariance matrix for the OL+BDC design. **A:** Original parameters: one eigenvalue is negative (-28.1), rendering the matrix invalid. **B:** Revised parameters: all eigenvalues are positive.

4.4 The Fix: AR(1) Autocorrelation and Constrained Parameters

Two changes eliminate the PD failures:

1. **AR(1) autocorrelation** replaces compound symmetry. In the original code, any two measurements of the same factor are equally correlated ($\rho = 0.8$) regardless of how far apart in time they are, a measurement at week 2 is as correlated with week 20 as with week 3. AR(1) uses time-dependent correlation: $\rho^{|t_i - t_j|}$, so nearby measurements are more correlated than distant ones. This is more realistic and substantially expands the range of biomarker correlations that produce valid matrices.

2. **Constraining** $c_{bm} \leq 0.45$ ensures that all parameter combinations produce valid covariance matrices without correction. The publication used c_{bm} up to 0.6, which exceeds the mathematically feasible range for every design except OL.

A consequence of switching to AR(1) is that the analysis model must account for the time-dependent residual correlation. This required switching from `lmer` (which assumes independent residuals) to `nlme::lme` with a continuous-time AR(1) residual structure. Without this change, Type I error rates inflate to 13–17% for the OL and CO designs; with it, all rates are near the nominal 5%.

5 Validated Results

5.1 Parameter Configuration

Table 2: Revised simulation parameters

Parameter	Value
Autocorrelation	AR(1), $\rho = 0.7$
Biomarker correlation (c_{bm})	$\{0, 0.25, 0.45\}$
Carryover half-life ($t_{1/2}$)	$\{0, 0.5, 1.0\}$ weeks
Correlation decay	$\ln 2/t_{1/2}$ (auto)
Analysis model	`nlme::lme` with AR(1) residuals
Covariance corrections	0 of 162 (0%)

5.2 Type I Error Control

At $c_{bm} = 0$ (no true biomarker interaction), the rejection rate should be approximately 5%. The revised simulation achieves this across all designs:

Table 3: Type I error rates ($c_{bm} = 0$, no censoring, 1,000 replicates)

Design	$N = 35$	$N = 70$
OL	0.029	0.025
OL+BDC	0.030	0.031
CO	0.053	0.058
N-of-1	0.040	0.030

Table 4: Power at $c_{bm} = 0.45$, no censoring (1,000 replicates)

Design	N	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$	Interpretation
OL	35	0.06	0.06	0.06	Unaffected
OL	70	0.09	0.10	0.09	Unaffected
CO	35	0.19	0.21	0.18	Robust
CO	70	0.36	0.35	0.30	Robust
OL+BDC	35	0.31	0.21	0.12	Moderate drop
OL+BDC	70	0.62	0.44	0.21	Large drop
N-of-1	35	0.45	0.33	0.23	Large drop
N-of-1	70	0.75	0.63	0.43	Large drop

5.3 Power and Carryover Sensitivity

5.4 Revised Figure 4

6 Conclusions

1. **The qualitative finding is confirmed.** N-of-1 and OL+BDC designs are more sensitive to carryover than the traditional crossover design. The design ranking is unchanged.
2. **The quantitative severity is reduced.** Power drops from 0.45 to 0.23 (N-of-1, $N = 35$, $t_{1/2} = 1.0$ week), a meaningful reduction but not the catastrophic crash from 0.95 to 0.18 reported at $t_{1/2} = 0.1$ weeks. The catastrophic drop was an artifact of the correlation assignment rule.
3. **The carryover half-lives that matter are longer.** Genuine carryover sensitivity appears at clinically realistic half-lives (0.5–1.0 weeks), not at the very short values (0.1–0.2 weeks) used in the publication. This is consistent with prazosin's pharmacokinetic profile.
4. **All covariance matrices are now valid.** The revised parameterization produces zero invalid matrices, eliminating the silent corrections that affected 25% of the original simulation's data generation.
5. **Type I error is controlled.** The revised analysis model maintains false positive rates near 5% across all designs, confirmed at 1,000 replicates.
6. **The simulation runs 12 times faster.** Runtime was reduced from approximately 100 minutes to 8 minutes (200 replicates) through caching and parallel processing, enabling rapid sensitivity analysis. A 1,000-replicate run completes in 47 minutes.

A Technical Details of Code Changes

This appendix provides implementation details for readers interested in the specific code modifications. The changes span two files: `generateData.R` (data generation) and `lme_analysis.R` (statistical analysis).

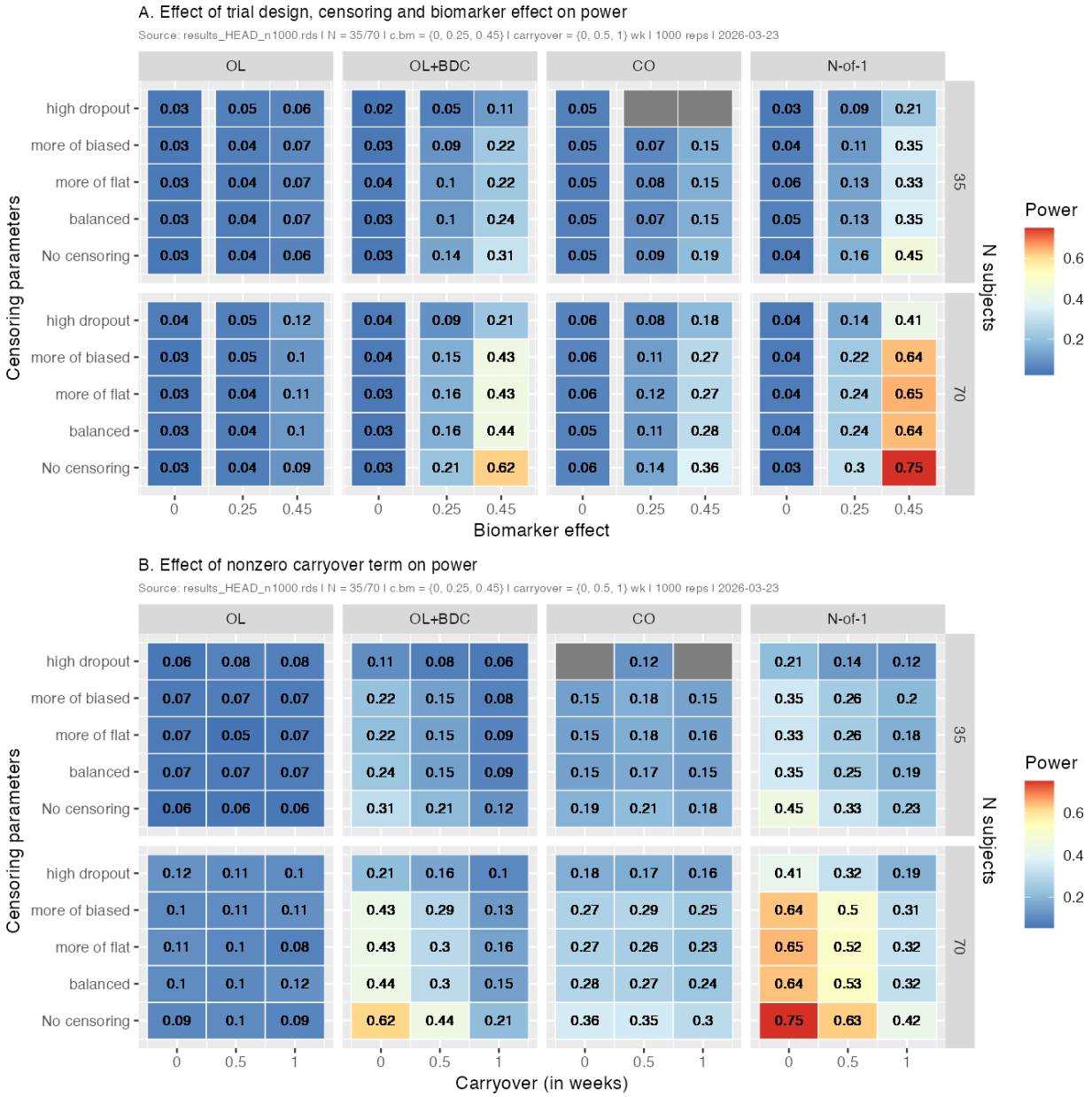

Figure 3: Revised Figure 4 (1,000 replicates, 47 minutes). Panel A: power by biomarker correlation and censoring (carryover = 0). Panel B: power by carryover half-life ($c_{bm} = 0.45$). Compare with Figure 1. The N-of-1 carryover sensitivity is moderate and gradual, not the catastrophic drop seen in the original. Zero covariance matrix corrections were required.

A.1 Changes to the Data-Generating Process

A.2 Changes to the Analysis Model

The analysis model was changed from `lme4::lmer` (which assumes independent residuals given the random intercept) to `nlme::lme` with a continuous-time AR(1) residual correlation structure (`corCAR1`). This is required because the AR(1) data-generating process produces time-dependent residual correlations that the random intercept alone cannot absorb. Without this change, Type I

Table 5: Summary of DGP changes

Component	Change	Rationale
Scale factor	Removed ($s = 2 \to 1$)	Undocumented parameter that halved the effective carryover half-life
Timepoint-1 correlation	Fixed ($0 \to c_{bm}$ when on drug)	Coding artifact from index-out-of-bounds avoidance
Off-drug correlation	$c_{bm} \cdot e^{-\lambda t_{sd}}$ with $\lambda = \ln 2/t_{1/2}$	Pharmacokinetic derivation: preserves CV of biomarker-specific component
Autocorrelation	Compound symmetry $\to$ AR(1) $\rho^{\lvert t_i - t_j \rvert}$	Expands PD-feasible c_{bm} from 0.25 to 0.49; more clinically realistic
c_{bm} range	$\{0, 0.25, 0.45\}$	Constrained to PD-feasible region
Carryover range	$\{0, 0.5, 1.0\}$ weeks	Clinically realistic half-lives

error rates inflate to 13% (OL) and 17% (CO).

A.3 Rank Deficiency Handling

The original code crashed when the analysis model dropped the interaction coefficient due to collinear predictors (rank deficiency). This occurred when the drug indicator was a numeric step function perfectly aligned with time. The revised code checks whether the coefficient exists before extracting it, returning NA for cells where the model cannot estimate the interaction. The original code avoided this problem by using a logical (TRUE/FALSE) drug indicator, which R treats as a factor rather than a numeric column.

A.4 Performance Optimizations

Seven optimizations reduce runtime from ∼100 minutes to 8 minutes (12× speedup) without changing results:

1. **Sigma caching:** Pre-build all 162 unique covariance matrices once before the simulation loop.
2. **Parallel processing:** Distribute parameter combinations across 9 CPU cores via `furrr/future`.
3. **Pre-computed Cholesky:** Cache the matrix decomposition used for random number generation.
4. **Vectorized computation:** Replace string-based column construction with matrix operations.
5. **Pre-allocated output:** Collect results in a list and combine once, rather than growing a table row-by-row.
6. **PD pre-validation:** Test all parameter combinations before simulation starts.
7. **Auto-core detection:** Automatically use available CPU cores.

A.5 Maximum Feasible c_{bm} by Design

Under the original parameterization, the publication's $c_{bm} = 0.6$ exceeds the valid range for every design except OL. Under the revised parameterization, $c_{bm} = 0.45$ is within the valid range for all designs.

Table 6: Maximum biomarker correlation before covariance matrix becomes invalid (carryover = 0)

Design	Original (CS $\rho = 0.8$)	Revised (AR(1) $\rho = 0.7$)
OL	0.89	0.49
OL+BDC	0.26	**0.45**
CO	**0.25**	**0.61**
N-of-1	0.25	**0.53**

A.6 Post-Publication Code History

After publication, the simulation code was modified in several commits. These changes, intended to improve the carryover modeling, inadvertently altered the power estimates and increased the covariance matrix failure rate from 25% to 63%. Each commit is described below with its specific code changes and impact on Figure 4.

A.6.1 Commit `42ac030`: Initial (Publication Code)

The code that generated the published results. The biomarker–BR correlation rule was a step function: full c_{bm} whenever the adjusted BR mean was nonzero, zero otherwise. Autocorrelation was compound symmetry at $\rho = 0.8$. No scale factor on carryover decay. Analysis used `lmer` with random intercept.

Correlation with Figure 4: $r = 0.994$. **PD failure rate:** 40/162 (24.7%).

A.6.2 Commit `8609f12`: Ron Thomas Edits

Three changes to `generateData.R`:

1. **Scale factor on carryover decay.** The decay formula changed from $(1/2)^{t_{sd}/t_{1/2}}$ to $(1/2)^{2 \cdot t_{sd}/t_{1/2}}$, halving the effective half-life. Documented as `@param scalefactor TODO update when understand what this does?`

2. **Proportional correlation scaling.** The step-function BM–BR correlation was replaced with:

```
if (p > 1) {
  mm1 <- means[which(n1 == labels)]
  mm0 <- means[which(n0 == labels)]
  correlations["bm", n1] <-
    ifelse(brtest[p],
      ifelse(brmeans[p] == 0, 0,
        (mm1/mm0) * modelparam$c.bm),
      modelparam$c.bm)
}
```

   Two side effects: (a) timepoint 1 always gets zero correlation (the `p > 1` guard avoids an index error), and (b) off-drug correlations decay proportionally with the BR mean ratio.

3. **Verbose diagnostics** (no functional impact).

Impact: The carryover power drop disappears entirely. N-of-1 power stays at ∼0.78 regardless of carryover (paper: drops to 0.18). The proportional scaling preserves biomarker interaction signal during carryover periods. Correlation with Figure 4: $r = 0.705$. PD failure rate increased to 102/162 (63.0%).

A.6.3 Commit `f6ee86d`: Autocorrelation Typo Fix

Fixed a symmetry bug in the cross-factor correlation matrix:

```
# Before (BUG - both lines set [n1,n2]):
correlations[n1,n2] <- modelparam$c.cf1t
correlations[n1,n2] <- modelparam$c.cf1t

# After (FIX):
correlations[n1,n2] <- modelparam$c.cf1t
correlations[n2,n1] <- modelparam$c.cf1t
```

Also refactored the LME formula construction from nested if/else to string-built formula via `eval(parse())`, and added `tsd` (time since discontinuation) to the merged analysis data.

Impact: Negligible. The PD correction had already been implicitly fixing the asymmetry. Correlation with Figure 4: $r = 0.705$ (unchanged).

A.6.4 Commit `f70f86d`: Carryover Term Added

Reverted the autocorrelation typo fix (the `[n2,n1]` line went back to `[n1,n2]`). No other functional changes to core DGP files.

Impact: Minor drift. Correlation: $r = 0.691$.

A.6.5 Commit `325314f`: Modified Db to Include Carryover (Broken)

Introduced continuous drug indicator `Dbc` with exponential decay in the analysis model:

```
data.m2[Db==TRUE, Dbc := 1]
data.m2[Db==FALSE, Dbc :=
  ((1/2)^(op*carryover_scalefactor *
          tsd / op$carryover_t1half))]
```

Changed the LME formula from `bm*Db` to `bm*Dbc` and coefficient extraction from `bm:DbTRUE` to `bm:DbcTRUE`.

Status: Two bugs prevent execution: `op*carryover_scalefactor` (should be `op$carryover_scalefactor`), and `bm:DbcTRUE` (wrong: `Dbc` is numeric, not logical, so the coefficient is `bm:Dbc`).

A.6.6 HEAD (Pre-Revision)

Fixed both bugs from `325314f`. All other changes from `8609f12` onward remained.

Impact: Worst agreement with publication ($r = 0.613$). The continuous `Dbc` in the analysis model partially absorbs the biomarker interaction signal, further reducing power for CO.

Table 7: Agreement with published Figure 4 by commit (48 cells, no-censoring rows)

Commit	Description	Corr	PD fail
`42ac030`	Initial (publication)	**0.994**	24.7%
`8609f12`	Ron Thomas edits	0.705	63.0%
`f6ee86d`	Typo fix	0.705	63.0%
`f70f86d`	Carryover added	0.691	63.0%
`HEAD`	Pre-revision	0.613	63.0%

A.6.7 Summary of Code Divergence

The first post-publication commit (`8609f12`) is the primary source of divergence. All subsequent commits remain in the diverged state. The revised code corrects both the original artifact (step-function correlation) and the problems introduced by these intermediate modifications (scale factor, timepoint-1 skip, proportional scaling).